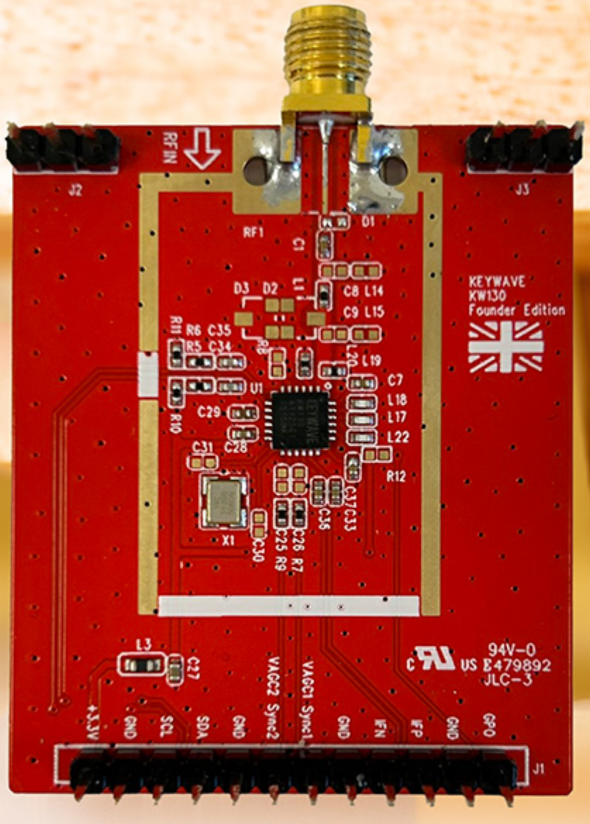
KEYWAVE



Keywave -

KW130

SDR Tuner

Datasheet

Revision 1.22
12-May-2025

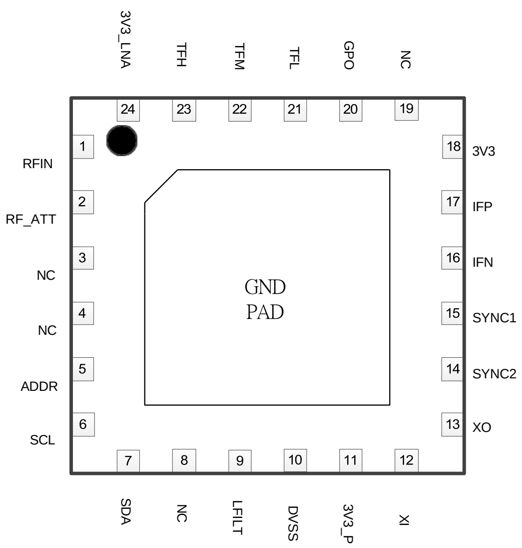
General Description

The KW130, a worldwide software design radio silicon tuner. This advanced tuner eliminates the need for external Baluns, LNAs, and SAW filters. Featuring an exceptional low-noise amplifier, smart detector, automatic gain control, and selective channel functionality, the KW130 excels in linearity, sensitivity, image rejection, and adjacent channel immunity.

Offering low noise, low spurious emissions, and reduced BOM costs, the KW130 streamlines design processes and enhances performance, making it an ideal solution for SDR(Software Defined Radio) receiver.

Applications

- Software defined radio receiver
- Analog/Digital TV
- Listening to aircraft traffic
- Tracking aircraft positions
- Listening to FM/RDS radio
- Spectrum Analyzer



Key Features

- Operation Frequency Range 30MHz to 2600MHz
- Support multi- channel band width up to 20MHz
- Low power consumption with 3.3V supply
- LO Sync function for muti-receiver phase sync.
- I^2C control interface
- 4x4 QFN package
- RoHS compliant

Revision	Date	Author
1.22	5/12/2025	Cloudio Tang

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Keywave KW130

Block Diagram

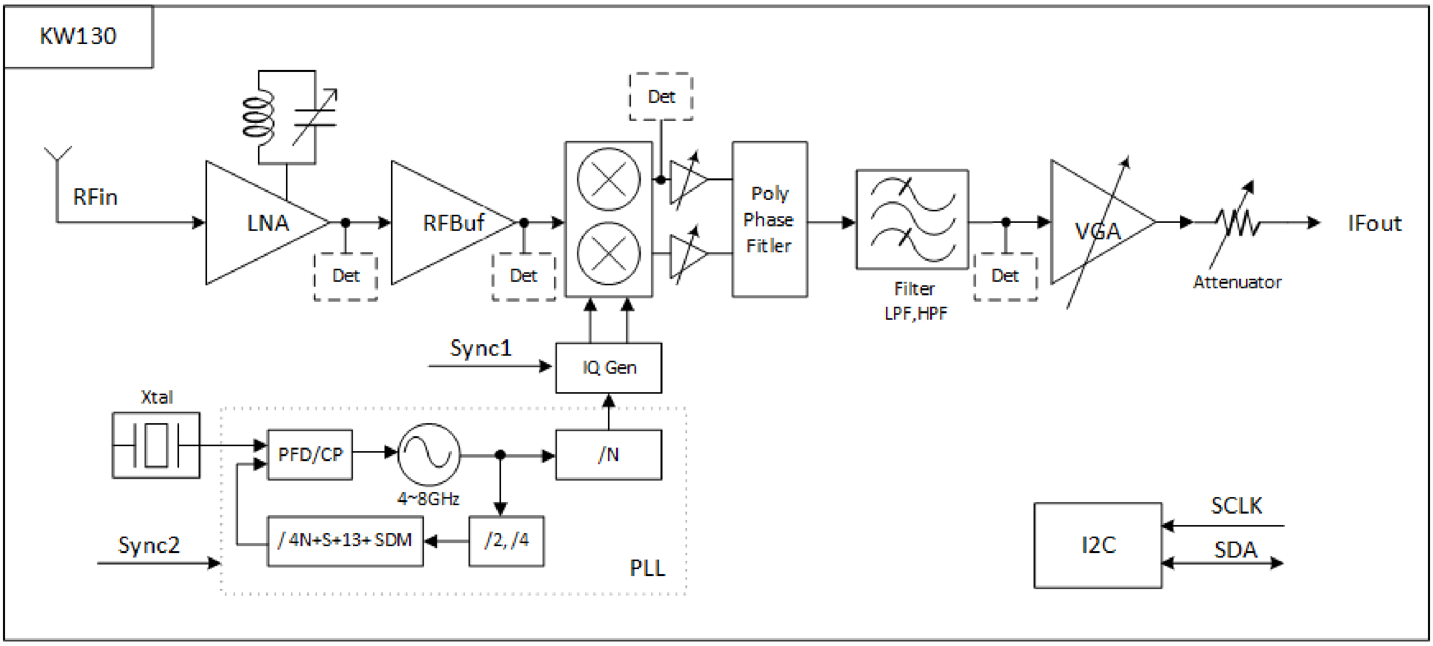


Figure 1: KW130 Block Diagram

Pin Information

Pin Assignments

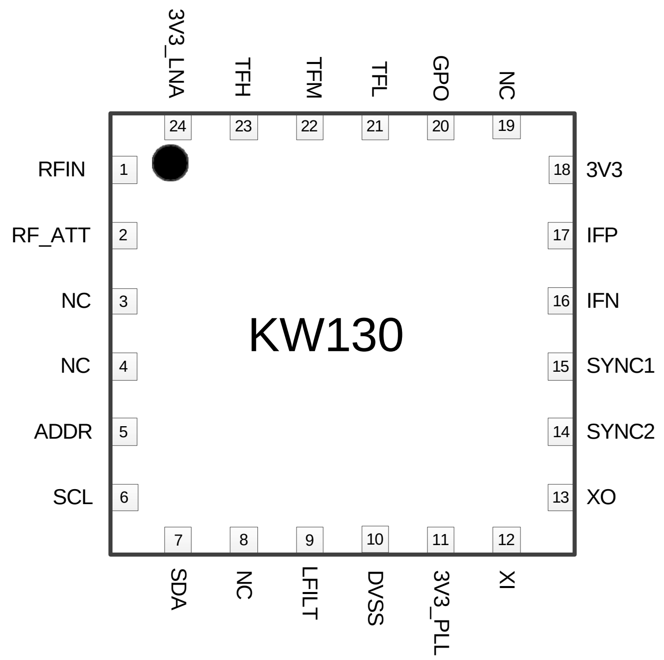


Figure 2:
KW130 Pin Assignments

Pin Descriptions

Pin	Symbol	Type	Description
1	RFIN	Input	RF input
2	RF_ATT	-	RF Attenuation Filter
3	NC	-	-
4	NC	-	-
5	ADDR	Input	Address selections (reference "KW130 Write/Read Address" of page 5)
6	SCL	Input	I^2C Clock
7	SDA	Bi-Directonal	I^2C Data
8	NC	-	-
9	LFILT	Input	PLL loop filter
10	DVSS	G	Ground for Digital blocks
11	3V3_PLL	Power	Supply voltage for PLL
12	XI	Input	Crystal input
13	XO	Output	Crystal output
14	SYNC2	Input	LO Sync control pin 2
15	SYNC1	Input	LO Sync control pin 1
16	IFN	Output	IF negative output
17	IFP	Output	IF positive output
18	3V3	Power	Supply Voltage for IF
19	NC	-	-
20	GPO	Output	General Purpose Output
21	TFL	-	LNA Tracking filter
22	TFM	-	LNA Tracking filter
23	TFH	-	LNA Tracking filter
24	3V3_LNA	Power	Supply Voltage for LNA

Electrical Specifications

Recommend Operating Conditions

Characteristic	Symbol	Min	Typical	Max	UNITS
Supply Voltage	VCC	3.0	3.3	3.6	V
Ambient Temperature	TA	-25	25	85	°C

Absolute Maximum Ratings

Characteristic	Symbol	Min	Max	UNITS
Supply Voltage	VCC	-0.3	3.6	V
AGC1/ AGC2 Control Voltage	VAGC	-0.3	3.6	V
I^2C input Voltage	VI^2C	-0.3	3.6	V

DC Characteristics

Characteristic	Symbol	Min	Typical	Max	UNITS
Supply Voltage	VCC	3.0	3.3	3.6	V
Current Consumption_ Operation (VDD=3.3v)	ICC_operation		180		mA
Current Consumption_ Standby (VDD=3.3v)	ICC_standby	-	4	-	mA

Thermal Characteristics

Characteristic	Symbol	Min	Typical	Max	UNITS
Thermal Resistance (Junction to Ambient)	θ_{JA}	-	4.7	-	°C/W
Thermal Resistance (Junction to case)	θ_{JC}	-	4.5	-	°C/W
Maximum Junction Temperature	T_J	-	125	-	°C
Operation Casing Temperature (based on 25°C operating of Ambient Temperature)	T_C	-	45	-	°C
Maximum Ambient Temperature	T_A	-	100	-	°C
Operation Temperature (Ambient)	T_{OP}	-25	-	85	°C
Storage Temperature (Ambient)	T_{STG}	-65	-	150	°C
Based on 2 layers PCB					

Electrical Specifications (cont.)

RF Characteristics

Characteristic	Symbol	Min	Typical	Max	UNITS
Operation Frequency Range	f_{RF}	30	-	2600	dB
Input Return Loss (by 75ohm system)	S_{11}	-	-10	-8	dB
Noise figure (@ 660MHz)	NF		2.2		dB
RF gain range	AGC_{RF}	-	55	-	dB
IF Gain range	AGC_{IF}	-	42	-	dB
Image Rejection	IMR	-	-50	-	dB
Phase Noise_1KHz (LO 1G)	PN_{1KHz}	-	-96	-94	dB
Phase Noise_10KHz (LO 1G)	PN_{10KHz}	-	-103	-100	dB
Phase Noise_100KHz (LO 1G)	PN_{100KHz}	-	-104	-103	dB
IF Output Level	IF_{OUT}		-	2	Vp-p

Crystal Characteristics

Characteristic	Symbol	Min	Typical	Max	UNITS
Crystal Oscillator Frequency	FOSC	16	24	32	MHz
Frequency Accuracy	FACC	-	± 30	± 50	ppm
Sharing Crystal Input to X’tal in Level	VSHARE	0.5	-	3.3	Vp-p

I²C interface Descriptions

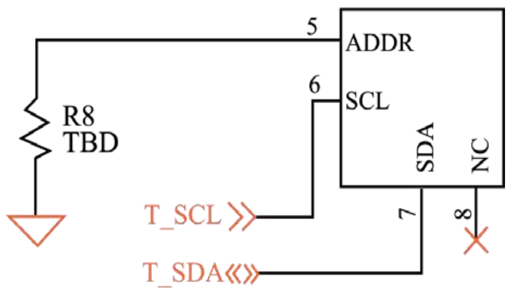
Communication between the KW130 and the host is facilitated through an I²C-compatible serial interface. The I²C uses SDA (Serial Data Line) and SCL (Serial Clock Line) for control and operates in both standard mode (100 kHz) and fast mode (400 kHz).

KW130 Write/Read Address

The KW130's addressing utilizes a fixed 7-bit slave address ID followed by a read/write (R/W) bit. The slave address ID is set to 1111001 for bits 7 through 1. The R/W bit is represented by "0" for write operations and "1" for read operations at bit 0. The formats for write and read operations are illustrated below:

Mode	I2C Address (Bin)							R/W	Address(hex)
	MSB	6	5	4	3	2	1	LSB	
Write	1	1	1	1	0	1	0	0	0xF2
Read	1	1	1	1	0	1	1	1	0xF3

Depending on the resistance of the R8 PIN, there are four different addresses.



R8_TBD			
0xF2 (Default)	0xB2	0x72	0x32
NC	75Kohm	22Kohm	0R

KW130 Write/Read Address (cont.)

K110 Write mode Write format is shown as follow:

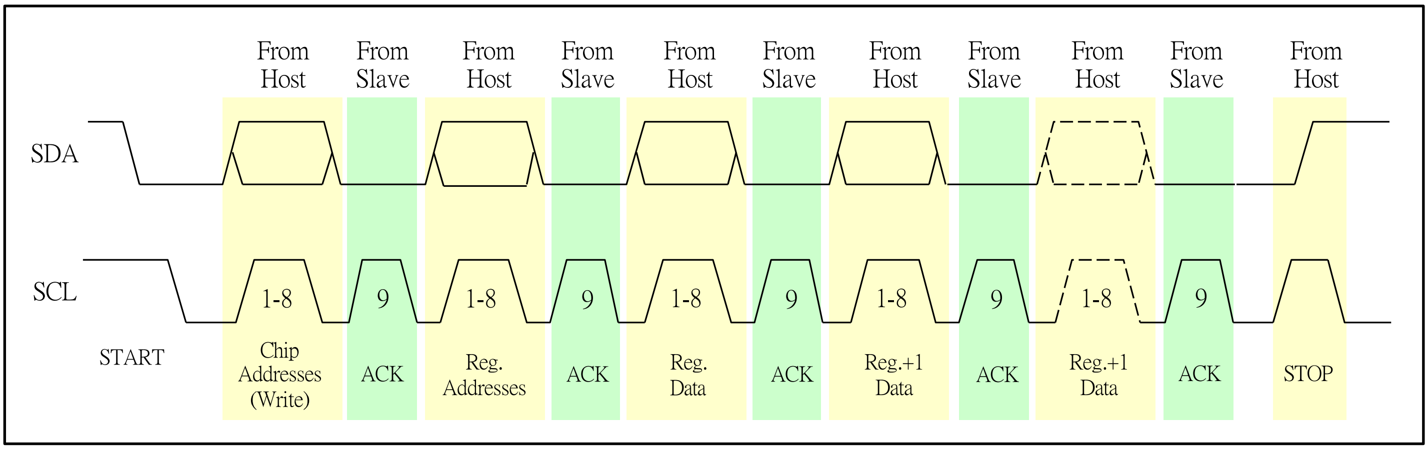


Figure 3: KW130 Write mode

K110 Read mode Read format is shown as follow:

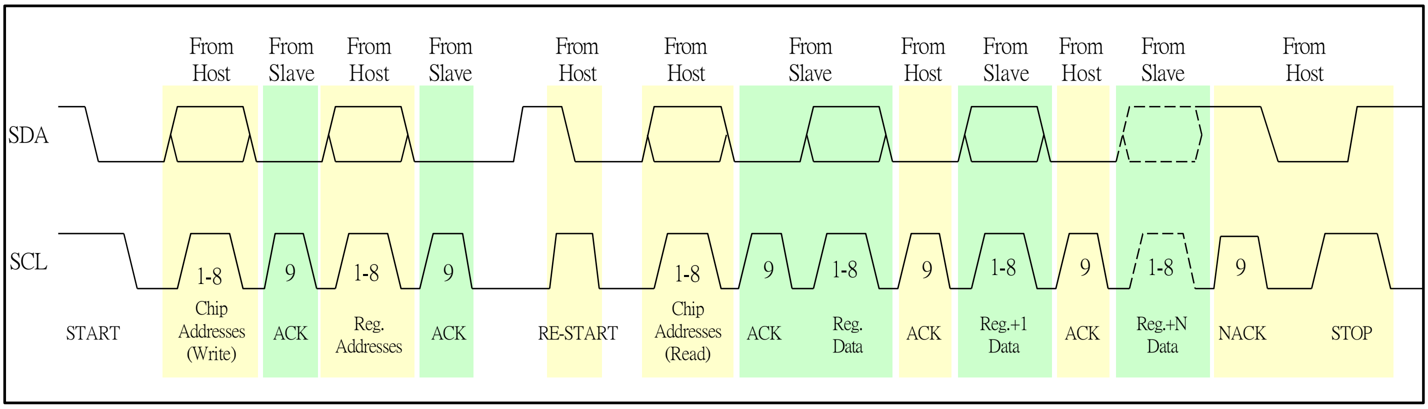


Figure 4: KW130 Read mode

I^2C interface Descriptions (cont.)

I^2C Electrical Characteristics

Parameters	Symbol	Condition	Units	Min	Max
Low level input voltage	V_{IL}	-	V	-0.5	0.2VDD
High level input voltage	V_{IH}	-	V	0.8VDD	3.8
Low level output voltage	V_{OL}	Pull up resistor of 4.7kohm to 3.3V	V	-	0.3
Capacitance for each I/O pin	C_i	-	pF	-	10

I^2C Timing Characteristics

The timing characteristics of the SDA and SCL for standard mode (100 kHz) and fast mode (400 kHz) are shown as follows:

Standard Mode					
Characteristic	**Symbol**	**Min**	**Typical**	**Max**	**UNITS**
Clock Frequency	f_{SCL}	0	-	100	kHz
SCL low time	t_{LOW}	4.7	-	-	µs
SCL High time	t_{HIGH}	4.0	-	-	µs
Setup time (repeated) START condition	t_{SU-STA}	4.7	-	-	µs
Hold time (repeated) START condition	t_{HD-STA}	4.0	-	-	µs
SDA Input to SCL Setup	t_{SU-DAT}	250	-	-	µs
SDA Hold Time	t_{HD-DAT}	0	-	3450	µs
Set-up time for STOP and START condition	t_{SU-STO}	4.0	-	-	µs
Bus free time between a STOP and START condition	t_{BUF}	4.7	-	-	µs
SDA Input SCL Rise/Fall Time	t_r t_f	-	-	1000	ns
Capacitive load for each bus line	C_b	-	-	30	pF
Input Filter Pulse Suppression	T_{sp}	-	-	50	ns

I2C interface Descriptions (cont.)

Fast Mode					
Characteristic	**Symbol**	**Min**	**Typical**	**Max**	**UNITS**
Clock Frequency	f_{SCL}	0	-	400	kHz
SCL low time	t_{LOW}	1.3	-	-	µs
SCL High time	t_{HIGH}	0.6	-	-	µs
Setup time (repeated) START condition	t_{SU-STA}	0.6	-	-	µs
Hold time (repeated) START condition	t_{HD-STA}	0.6	-	-	µs
SDA Input to SCL Setup	t_{HD-STA}	100	-	-	µs
SDA Hold Time	t_{HD-DAT}	0	-	900	µs
Set-up time for STOP and START condition	t_{SU-STO}	0.6	-	-	µs
Bus free time between a STOP and START condition	t_{BUF}	1.3	-	-	µs
SDA Input SCL Rise/Fall Time	t_r t_f	-	-	300	ns
Capacitive load for each bus line	C_b	-	-	30	pF
Input Filter Pulse Suppression	T_{sp}	-	-	50	ns

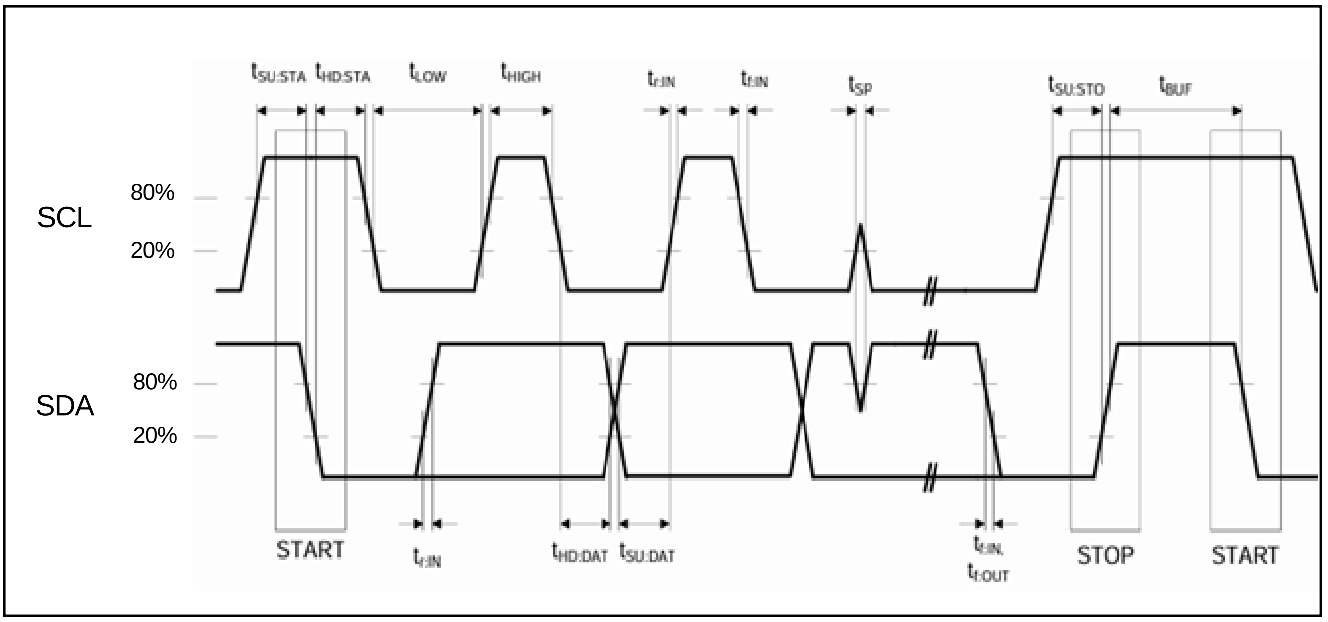


Figure 5: KW130 I2C Timing

Register Description

Table 3-1 lists KW130's register descriptions. For general case, user select opportune standard in KW130 driver, it does not need to manual setting and configure in all of registers.

Read Status	Register Bit	Description
Chip ID	R0[7:0]	0xB1
LNA AGC	R1[4:0]	LNA AGC Status, min: 0, max:31
Filter AGC	R1[7:5]	Filter AGC Status, min: 0, max:15
RF AGC	R2[3:0]	RF AGC Status, min: 0, max:15
Baseband AGC	R2[7:4]	Baseband AGC Status, min: 0, max:15
PLL VCO State	R3[7:0]	VCO Bank: 0 - 255
PLL Lock flag	R4[7]	1: PLL lock 0: PLL unlock

LNA Status: Write	Register Bit	Description
LNA Gain Mode	R14[0]	1 : Manual gain 0 : Auto control
Gain Value	R21[4:0]	LNA gain value at manual mode, 0~31 LNA Gain; min gain by "0", max gain by "31"
TF Band	R16[5:4]	Tacking filter band selection UHF band by "0" HF band by "1" MF band by "2" LF band by "3"
TF Cap	R17[6:0]	Tracking Filter Cap: 0 - 127
pw0_lna	R15[1:0]	LNA 1st stage current setting
pw1_lna	R15[2:3]	LNA 2nd stage current setting

RF BUF **Status: Write**	**Register Bit**	**Description**
RF Buf Gain mode	R18[4]	1: Manual gain 0: Auto control
rf_gain	R18[3:0]	RF Buffer gain value@ manual mode, 0~15 RF buffer gain; min gain by “0”, max gain by “15”
pw0_rfbuf	R19[2]	rfbuf current setting 0
pw1_rfbuf	R19[4:3]	rfbuf current setting 1
Poly_rfmux	R16[7:6]	Rf filter band selection 0: 200M~1024M 1: 130~200M, 2: <200M, 3: > 1024M
Rf_LPF	R18[7:6]	RF low pass filter, suggest setting: 0: > 250M, 1: 170M~250M, 2: 130M~170M, 3: <130M

Mixeramp **Status: Write**	**Register Bit**	**Description**
Mixeramp Gain Mode	R53[0]	1: Manual gain 0: Auto control
mixeramp_gain	R52{7:4]	mixeramp gain setting
Mixeramp_LPF	R6[3:2]	Mixeramp Low pass filter corner adjustment 0: lowest, 3: widest

Filter Status: Write	Register Bit	Description
Filter Gain Mode	R53[4]	Manual mode filter gain
Filt_gain	R53[3:1]	Filter gain value@ manual mode, 0~7 Filter gain; min gain by “0”, max gain by “7”
LPF_BW_S	R6[1:0]	LPF Filter Corner large scale adjust 0: < 9M 1: X 2: for 12M 3. for 16M/20M
IF_CH_BW	R41[7:5]	IF channel bandwidth middle scale setting
LPF_BW	R41[4:0]	IF LPF filter bandwidth fine setting 0~31 Narrowest by “0”; Widest by “31”
HPF_BW	R42[0:3]	High pass filter corner setting
eq_1th_LPF	R56[4:3]	1^{st} LPF equalization 0~3
eq_2th_LPF	R56[6:5]	2^{nd} LPF equalization 0~3

VGA Status: Write	Register Bit	Description
vga_code	R47[3:0]	VGA gain value 0~15 3dB/step
Vga_att	R57[6:4]	VGA attenuator for ADC input swing adjust 0~7 -2.5dB/step
Vga_eq	R55[1:0]	VGA Equalizer for IF output gain flatness 0~3

IO Pin Control Status: Write	Register Bit	Description
GPO Status	R63[7]	GPO output mode voltage 0V output by "0"; 3V3 output by "1"
En_sync	R5[0]	Enable pll sync control

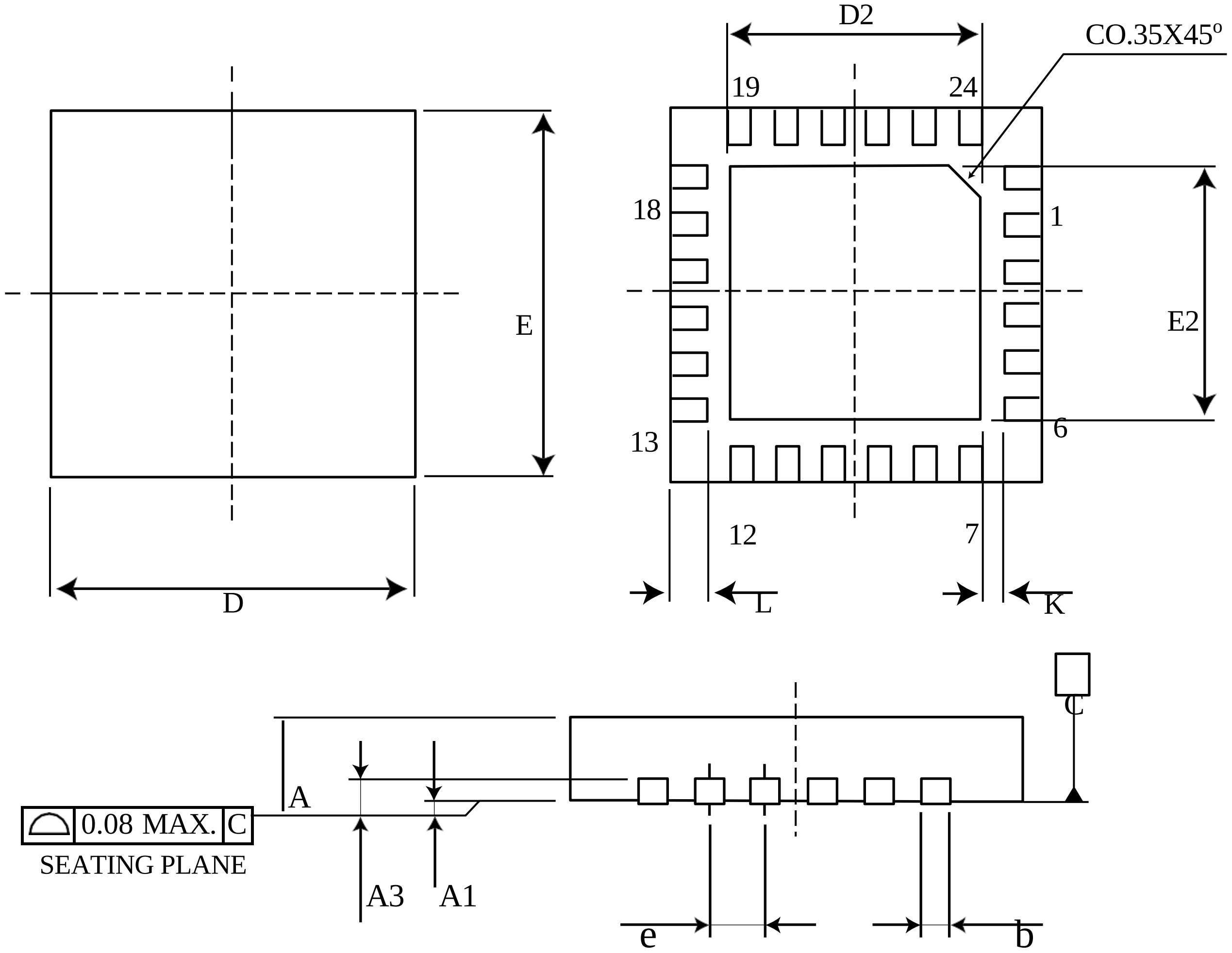
Register Description (AGC)

Gain Control (AGC) **Status: Write**	**Register Bit**	**Description**
LNA_VTL	R34[3:0]	LNA & RFBuf AGC threshold low
LNA_VTH	R34[7:4]	LNA & RFBuf AGC threshold high
MIX_VTL	R35[3:0]	mixeramp & filter AGC threshold low
MIX_VTH	R35[7:4]	mixeramp & filter AGC threshold high
clkf_agc	R36[1:0]	AGC clock (fast group) speed setting
clks_agc	R36[4:2]	AGC clock (slow group) speed setting
clkf_RC	R37[1:0]	AGC (fast group) RC corner for loop stability
clks_RC	R37[4:2]	AGC (slow group) RC corner for loop stability

TOP **Status: Write**	**Register Bit**	**Description**
TOP_LNA	R48[4:2]	Decide LNA AGC take off power point Value: min: 0 : max: 7
TOP_RFBuf	R48[7:5]	Decide BFBuf AGC take off power point Value: min: 0 : max: 7
TOP_Mixeramp	R49[2:0]	Decide Mixeramp AGC take off power point Value 0: Max Power Max-1 -7: Decreasing power levels from max
TOP_Filter	R49[5:3]	Decide Filter AGC take off power point Value 0: Max Power Max-1 -7: Decreasing power levels from max

Package Dimensions and Outline

KW130 is packaged by Lead-Free 4x4 24-pin Quad Flat No-Lead (QFN) package dimensions are shown.

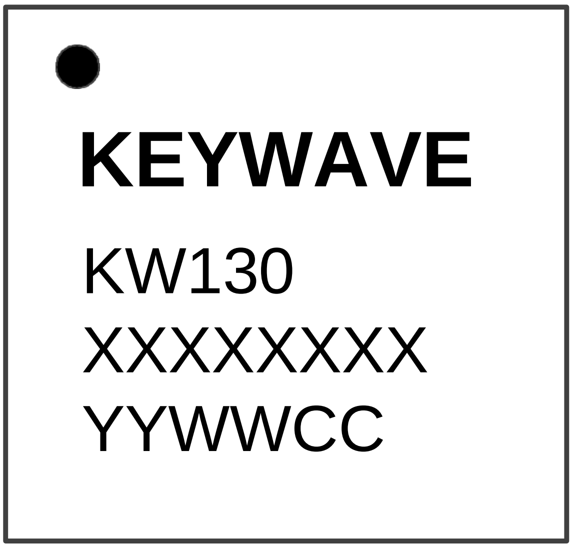


Package Dimensions

	PACKAGE TYPE		
JEDEC OUTLINE	MO-220		
PKG CODE	VQFN(Y424)		
SYMBOLS	MIN	NOM	MAX
A	0.8	0.85	0.9
A1	0.00	0.02	0.05
A3	0.20 REF		
b	0.18	0.25	0.30
D	4.00 BSC		
D2	2.60	2.70	2.75
E	4.00 BSC		
E2	2.60	2.70	2.75
L	0.35	0.40	0.45
e	0.50 BSC		
K	0.20	-	-

KW130 TOP Marking

TOP Marking



KW130 TOP Marking

Mark Method	Laser	
Font Size	Logo 3.1X1.3mm Device Number 0.45X0.3mm Mfg. Code& Date Code 0.45X0.3mm Spacing: 0.15mmX0.15mm	
Line 1 Marking	Circle=0.3 mm Diameter (Top-left justified) 0.5mmX0.5mm	Pin 1 identifier
Line 2 Marking	Logo	KEYWAVE
Line 3 Marking	Device Number	KW130
Line 4 Marking	(1) XXXXXXXX = Mfg. Code	Manufacturing Code from the Assembly Purchase Order form
Line 54 Marking	YYWWCC YY = Year/ WW = Work Week CC= Control Code	Assigned by the Assembly House. Corresponds to the year and work week of the mold date.

Ordering Information

Part Number	Shipping Package	Min. Order (MOQ)	MTQ

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